

Theory on React:

- Define SPA and its benefits.

A Single Page Application (SPA) is a web application that loads a single HTML page and dynamically updates its content without reloading the entire page. Only the required data is fetched from the server, making the application faster and more responsive.

Its benefits are:-

1. Faster page loading after the initial load.
2. Better user experience.
3. Reduced server requests.
4. Smooth navigation between pages.
5. Efficient use of bandwidth.

- Define React and identify its working.

React is an open-source JavaScript library developed by Meta (Facebook) for building interactive and reusable user interfaces, especially for Single Page Applications.

React works by breaking the UI into small reusable components and updating only the changed parts using the Virtual DOM.

- Identify the difference between SPA and MPA.

SPA	MPA
Loads a single HTML page	Loads a new page for new request
Faster after initial load	Slower due to page reloads
Better user experience	Full page refresh required
Uses JavaScript extensively	Server renders every page
Ex: Gmail , Facebook	Ex: Amazon , Traditional Websites

- Explain pros and cons of SPA.

Pros:-

1. Fast performance

2. Smooth navigation
3. Better user experience
4. Lower server load
5. Easy API integration

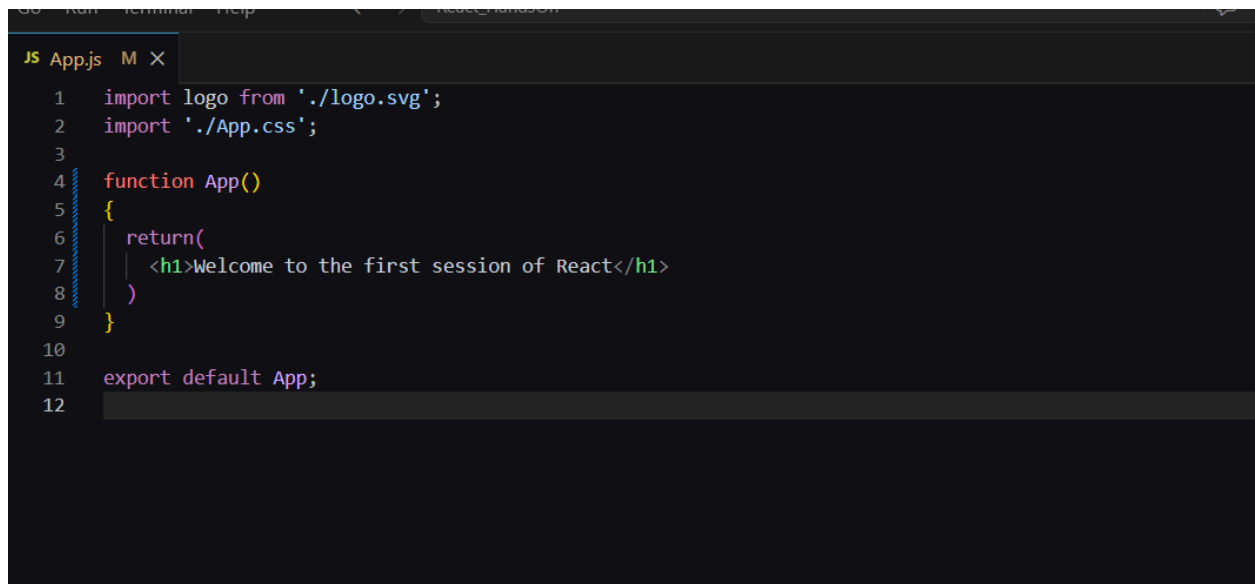
Cons:-

1. Initial loading time higher
 2. SEO is more difficult
 3. JavaScript dependency
 4. Browser history management can be complex
- Define Virtual DOM.

The Virtual DOM is a lightweight copy of the Real DOM. React first updates the Virtual DOM, compares it with the previous version using a diffing algorithm, and updates only the changed elements in the Real DOM. This makes React applications faster and more efficient.

- Explain features of React.
 1. Component-based Architecture: React applications are built using components, which are independent and reusable pieces of code. Each component manages its own functionality, making applications easier to develop, maintain, and reuse.
 2. Virtual DOM: React uses a Virtual DOM, a lightweight copy of the Real DOM. It compares changes in the Virtual DOM with the previous version and updates only the modified elements in the Real DOM, improving performance.
 3. JSX(JavaScript XML): JSX is a syntax extension that allows developers to write HTML-like code inside JavaScript. It makes React code more readable and easier to write while being converted into JavaScript by the compiler.
 4. One-Way Data Binding: React follows one-way (unidirectional) data flow, where data moves from parent components to child components through props. This makes applications easier to debug and ensures predictable behavior.
 5. Declarative UI: React uses a declarative programming approach, where developers describe how the UI should look for a given state. React automatically updates the interface whenever the data changes.

First Practical HandsOn:



A screenshot of a code editor window titled 'App.js'. The code is written in JavaScript and defines a function 'App()' that returns a JSX element. The code is as follows:

```
1 import logo from './logo.svg';
2 import './App.css';
3
4 function App()
5 {
6   return(
7     <h1>Welcome to the first session of React</h1>
8   )
9 }
10
11 export default App;
12
```

